

Identities in AKS

- **Managed identity**
 - Default identity type in AKS
 - Doesn't require credential authentication and provides automatic credential rotation
 - Azure provides two types of managed identities:
 - System-assigned
 - User-assigned
 - You can bring own identities, but AKS deploys automatically control plane identity as system-assigned and kubelet identity as user-assigned if you don't bring yours
 - Main responsibility of the kubelet identity is to authenticate with ACR
 - Other managed identity may be get created for specific addons that you enable during cluster creation or later, like Azure Monitor
- **Service Principal**
 - Security identity that represents an application or service and is used to authenticate and authorize access to Azure resources
 - Contains client ID and secret
 - The secret is by default valid for 1 year and then it needs rotation and AKS needs to be updated with the new value
 - Credentials are stored at node level in the file `/etc/kubernetes/azure.json`